

Assignment on Boolean Algebras

October 15, 2025

- Q1. 1. The glb operation ($\wedge$) and lub operation ($\vee$) are associative and commutative.
2. $x \vee 0 = x$, $x \wedge 0 = 0$, $x \vee 1 = 1$, and $x \wedge 1 = x$.
3. Any non-empty finite subset $\{x_1, \dots, x_k\}$ of B has a glb and a lub in B .
4. $x \leq y$, $u \leq v$ imply $x \vee u \leq y \vee v$ as well as $x \wedge u \leq y \wedge v$.
- Q2. 1. De Morgan's laws: $(x \wedge y)^c = x^c \vee y^c$; $(x \vee y)^c = x^c \wedge y^c$.
2. $x \leq y$ if and only if $y^c \leq x^c$.
3. $x \leq y^c$ if and only if $x \wedge y = 0$.
- Q3. In the power set Boolean algebra of a set U , viz. $(\mathcal{P}(U), \cup, \cap, ^c, \emptyset, U)$, the atoms are exactly the singletons.
The algebra is atomic.
- Q4. Consider the Boolean algebra $(\mathcal{F}/\equiv, \leq, \vee, \wedge, \neg, 0, 1)$, where $PV := \{p_1, \dots, p_n\}$.
1. For each n -tuple $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ in $\{0, 1\}^n$, $[\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i]$ is an atom.
2. Moreover, every atom in the algebra must be of this form.
3. The algebra is atomic.
- Q5. The Boolean algebra $(\mathcal{F}/\equiv, \leq, \vee, \wedge, \neg, 0, 1)$, where PV is *infinite*, is atomless.
- Q6. If a is an atom, then for all $x \in B$, either $a \leq x$ or $a \leq x^c$.
- Q7. For any homomorphism h from $\mathcal{B}$ to $\mathcal{B}'$, prove that
1. $h(x \vee y) = h(x) \vee h(y)$,

2. $h(0) = 0$ and

3. $h(1) = 1$.

Q8. Let h be an onto map from B to B' , where $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$, $\mathcal{B}' := (B', \leq, \vee, \wedge, ^c, 0, 1)$ are Boolean algebras. Show that h is an isomorphism from $\mathcal{B}$ to $\mathcal{B}'$ if and only if for all $x, y \in B$, $x \leq y$ if and only if $h(x) \leq h(y)$.

Q9. 1. Let h be a isomorphism between Boolean algebras $\mathcal{B}$ and $\mathcal{B}'$. Show that a is an atom in $\mathcal{B}$ if and only if $h(a)$ is an atom in $\mathcal{B}'$.

2. Let $\mathcal{B}$ and $\mathcal{B}'$ be isomorphic Boolean algebras. Prove that if one is atomic, the other is also so; if one is atomless, the other is also so.

Q10. Let $\mathcal{B}$ be a finite Boolean algebra. For every $x (\neq 0) \in B$, $x = a_1 \vee \dots \vee a_n$, where $\{a_1, \dots, a_n\}$ is the set of all atoms such that $a_i \leq x, i = 1, \dots, n$.

Q11. Consider the set S of all finite and *cofinite* subsets (i.e. those whose complements are finite) of an infinite set U .

1. Show that S forms a Boolean subalgebra of the power set Boolean algebra $(\mathcal{P}(U), \cup, \cap, ^c, \emptyset, U)$.

2. Show that the set of all cofinite subsets of U forms a filter (it is called the *Frechet filter* on U).

Q12. The *principal filter* $F(a)$ generated by any element a of a Boolean algebra $\mathcal{B}$ is defined as the smallest filter in $\mathcal{B}$ that contains a . Show the following.

1. $F(a) = \{x \in B : a \leq x\}$.

2. $F(a)$ is proper if $a \neq 0$.

3. $F(a)$ is prime if a is an atom.

Q13. For any prime filter F , there is a map $h : B \longrightarrow \{0, 1\}$ such that h is a homomorphism from $\mathcal{B}$ to $\mathbf{2}$, and $F = \{x \in B : h(x) = 1\}$.

Q14. Prove the converse of Q13.

Q15. Let $\mathcal{B}$ be a finite Boolean algebra. Prove that the set of atoms of $\mathcal{B}$ is in bijection with the set of all prime filters of $\mathcal{B}$.

You may also try out Questions 1, 3-7, 11, 18 from the exercise on Chapter 2 in Cori and Lascar. Assume in each case, the order-based definition of Boolean algebra (that we have learnt in class).